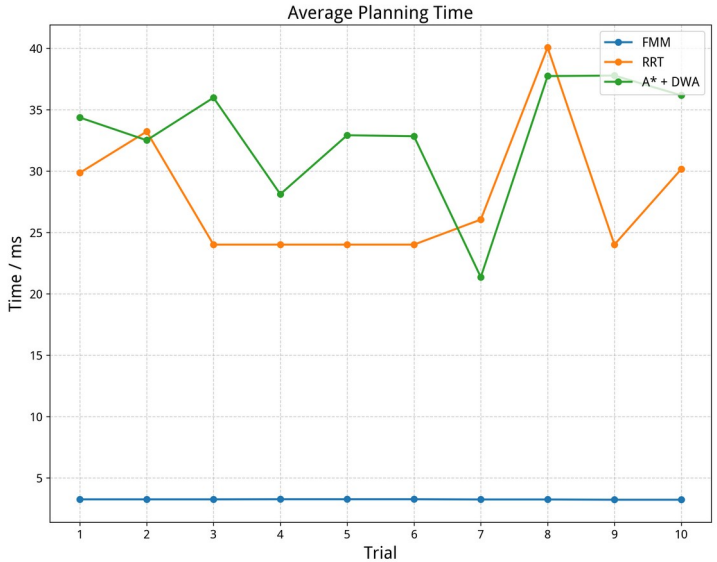
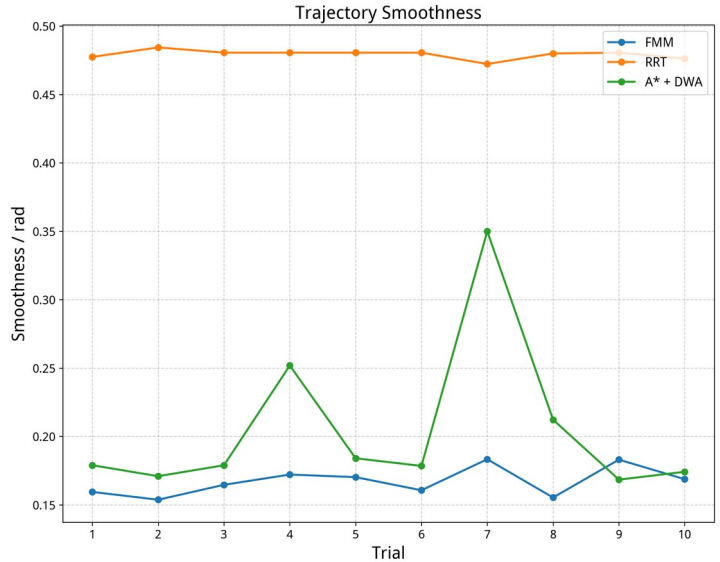
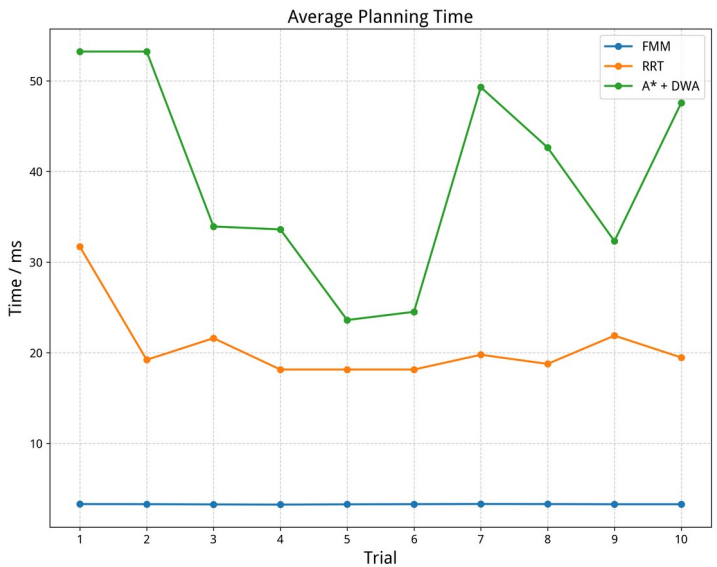

Average Planning Time
FMM
RRT
A* + DWA
Time / ms
Trial

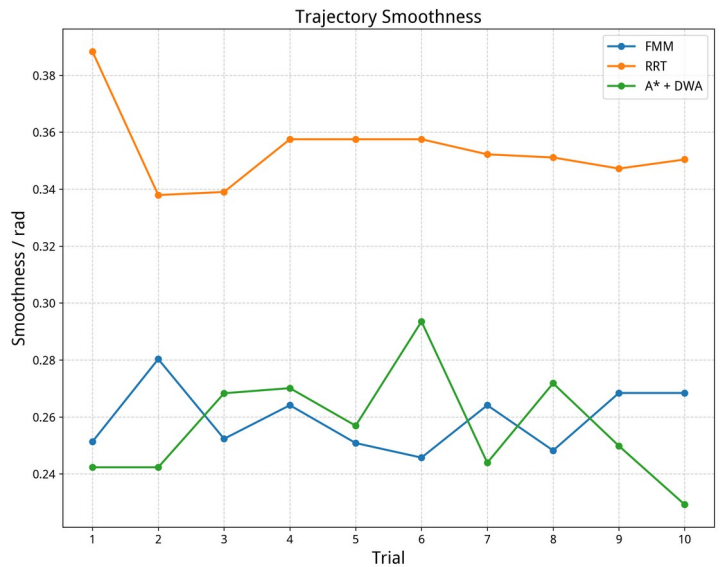
(a)


Trajectory Smoothness
FMM
RRT
A* + DWA
Smoothness / rad
Trial

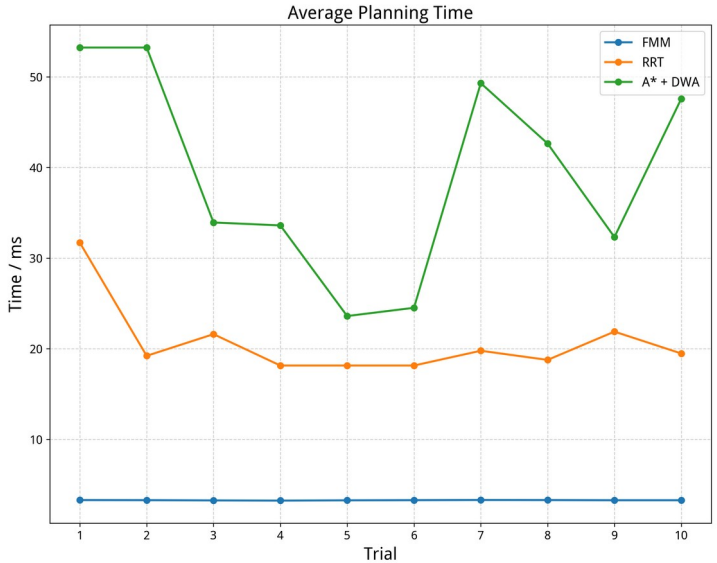
(b)


Average Planning Time
FMM
RRT
A* + DWA
Time / ms
Trial

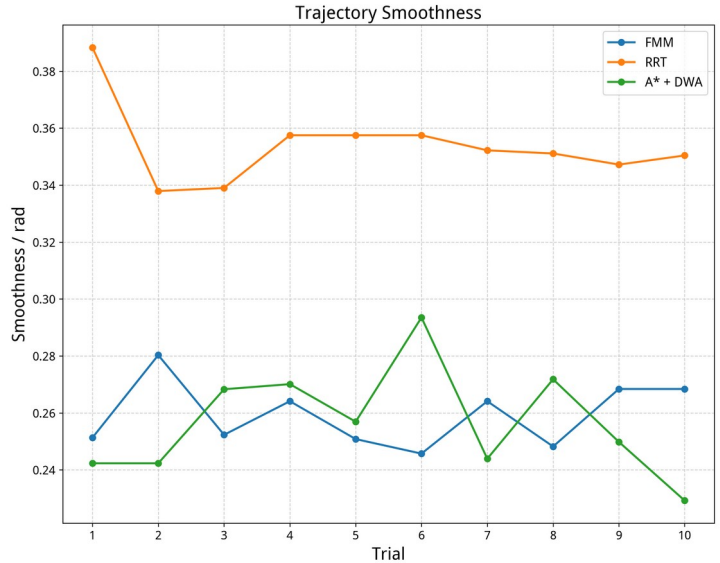
(a)


Trajectory Smoothness
FMM
RRT
A* + DWA
Smoothness / rad
Trial

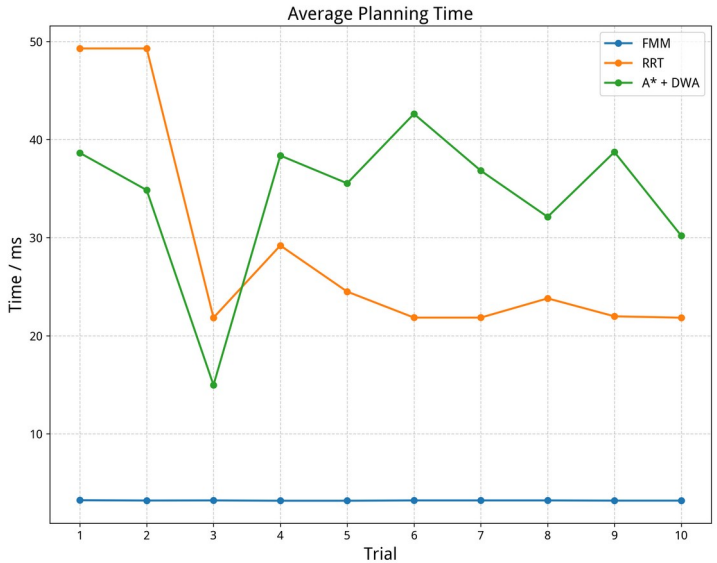
(b)


Average Planning Time
FMM
RRT
A* + DWA
Time / ms
Trial

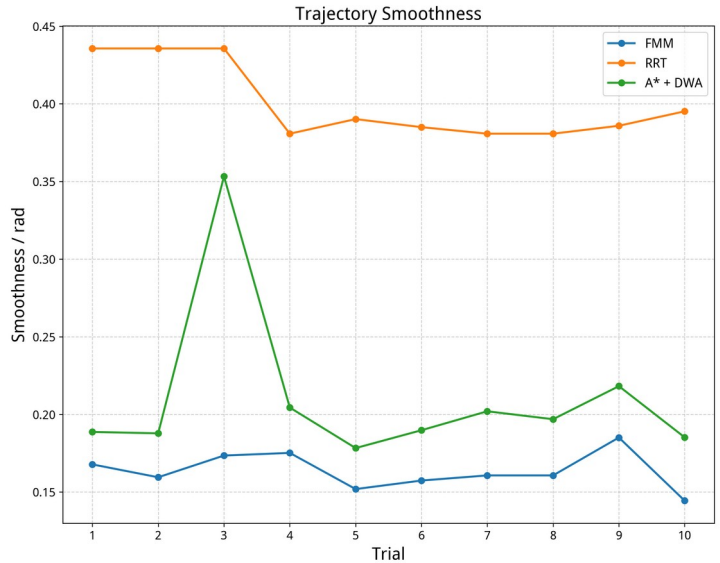
(a)


Trajectory Smoothness
FMM
RRT
A* + DWA
Smoothness / rad
Trial

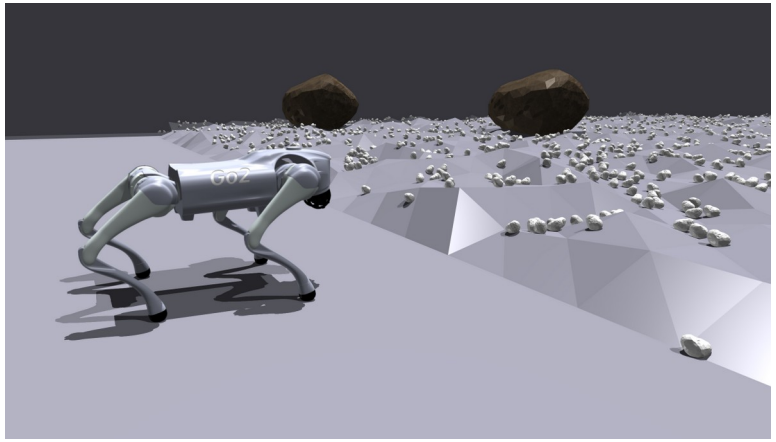
(b)


Average Planning Time
FMM
RRT
A* + DWA
Time / ms
Trial

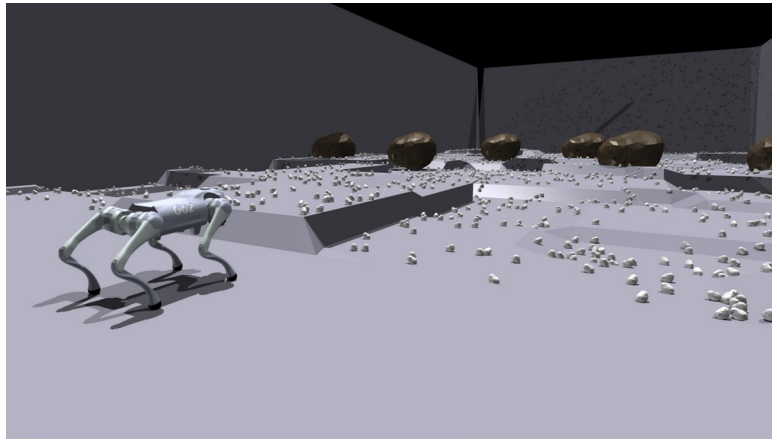
(a)


Trajectory Smoothness
FMM
RRT
A* + DWA
Smoothness / rad
Trial

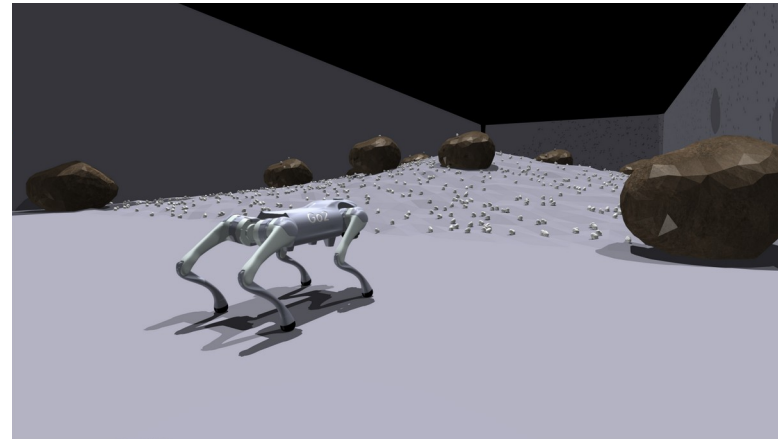
(b)



(a) Rough Slopes



(b) Discrete Obstacles



(c) Rough Flat





(a) Depth Image



(b) BEV Image

Hierarchical Visual Navigation and Locomotion Framework

Completed Chapter 3 system and evaluation pipeline

Perception Inputs





VIO Pose
(ideal or noisy)

Local Goal

Locomotion Inputs

Desired Base Velocity

Proprioceptive State

Historical Observations

Local Perception and Planning

RGB-D Projection
Point Cloud to BEV

Local BEV
Height / Obstacles

Dual-layer Cost
Hard + Soft Safety

FMM Reference
Corridor

Footprint MPPI
Trajectory
Optimization

Velocity Command
(vx, wz)

HIMLoco Locomotion Controller

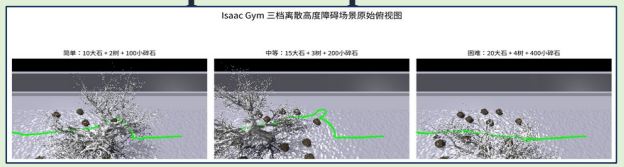
State and History
Encoder

HIMLoco
Policy Network

Joint Position
Targets

Shared locomotion policy for all navigation planners

Completed Experiments



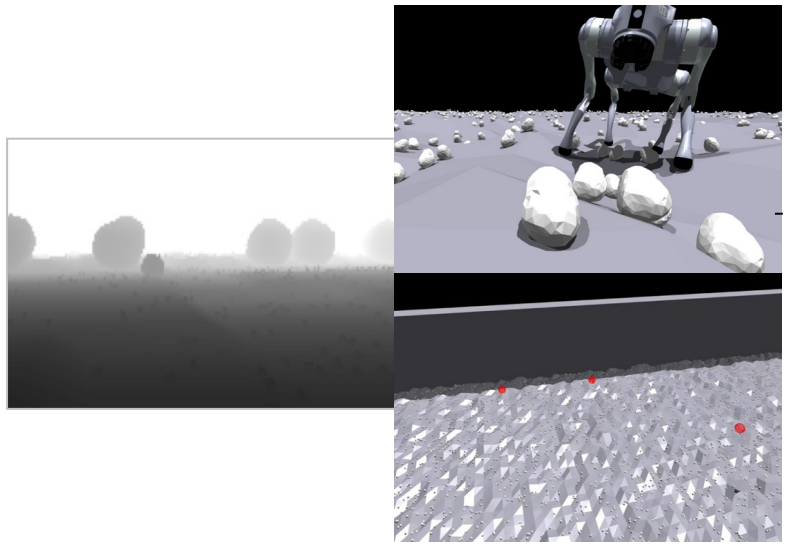
Three difficulty levels
30 fixed seeds per method

Internal ablation
Fixed FMM, Dual-layer FMM,
Footprint MPPI

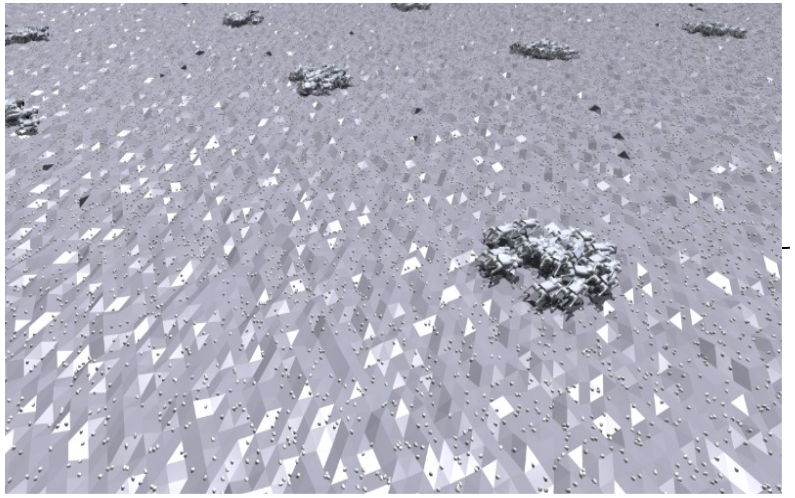
External comparison
iPlanner, PS-DWA, SAPF

VIO robustness
ideal, mild, medium, severe

750 formal closed-loop trials



Depth Camera Perception Driving



Reward Function Driving

Trajectory Planning

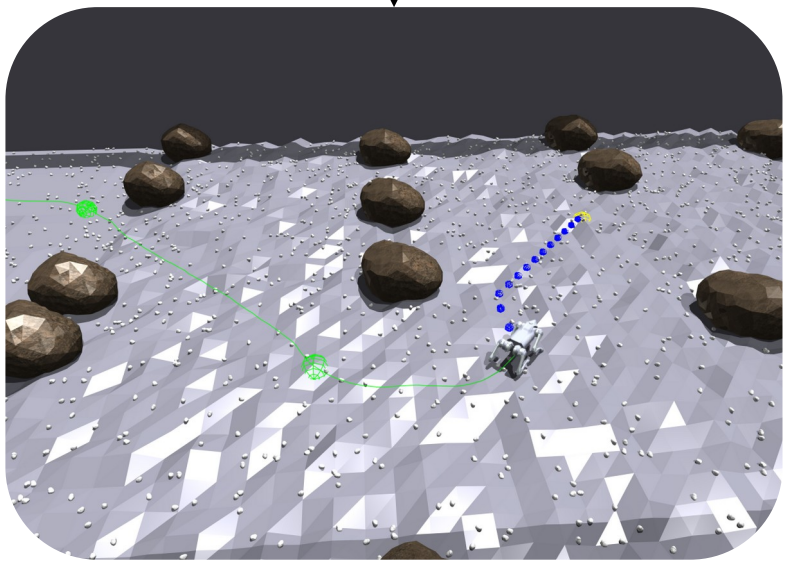
Reinforcement Learning

Navigation Controller

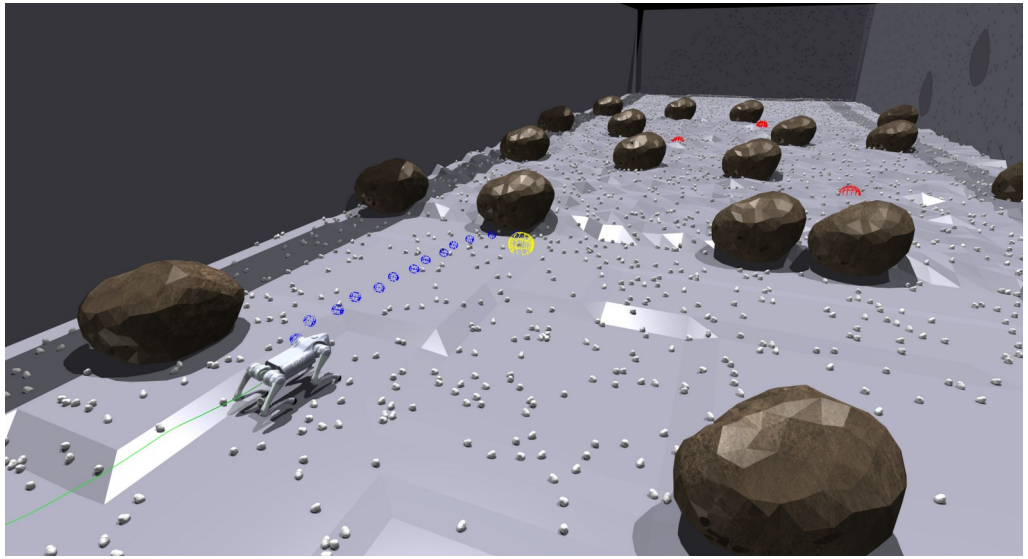
v_x, ω_{yaw}

Gait Controller

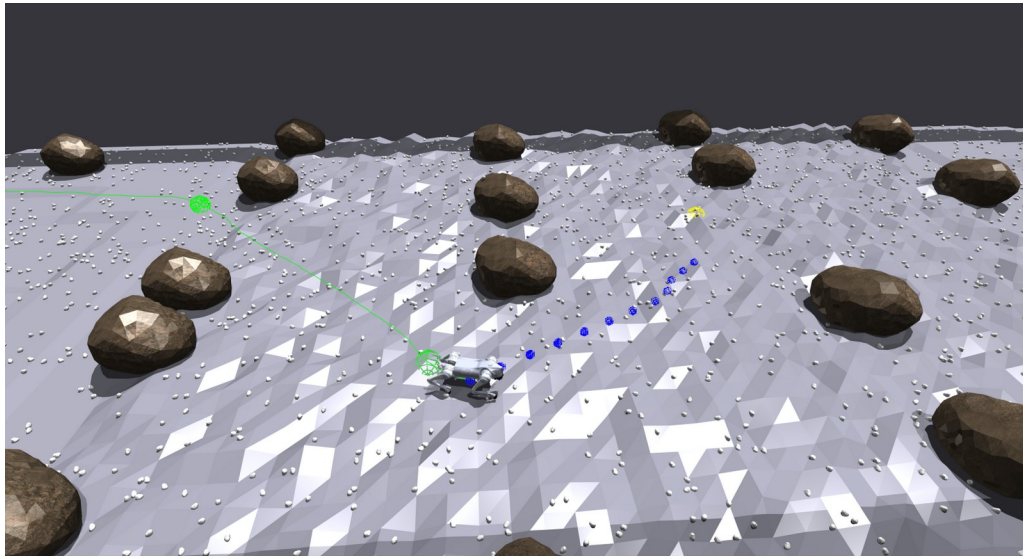
Joint Position Commands



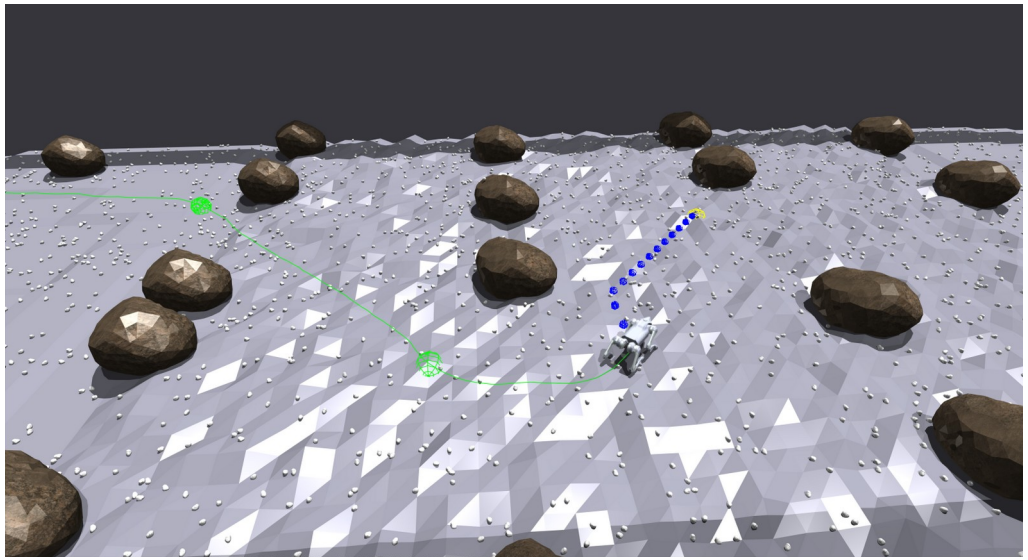
Autonomous Navigation



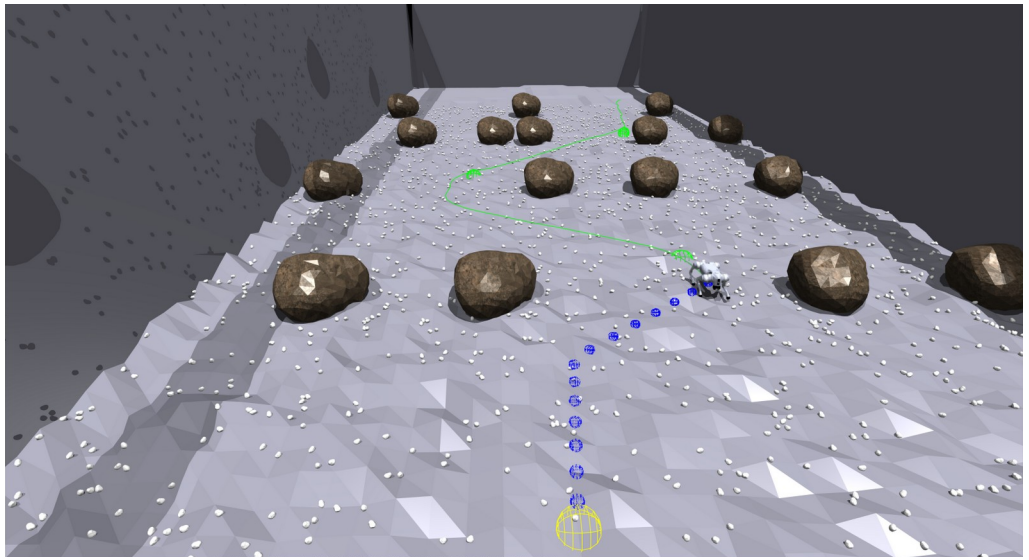
(a)Start Point — Waypoint A



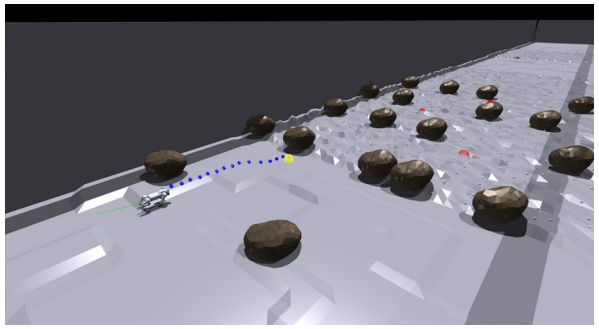
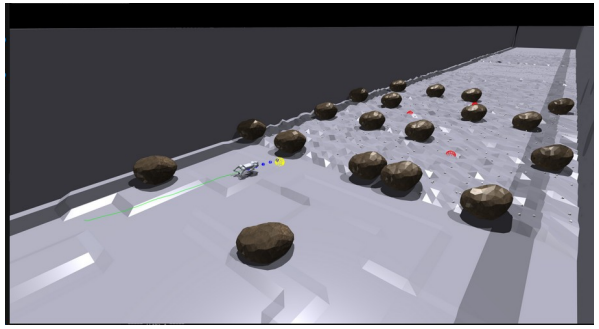
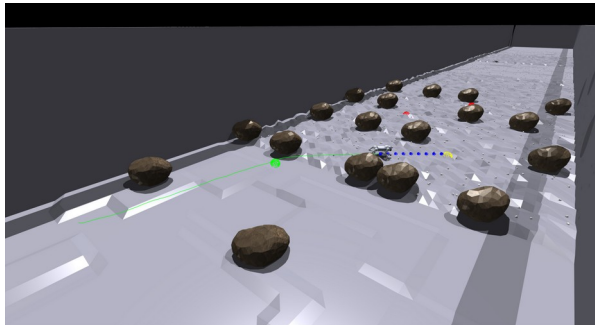
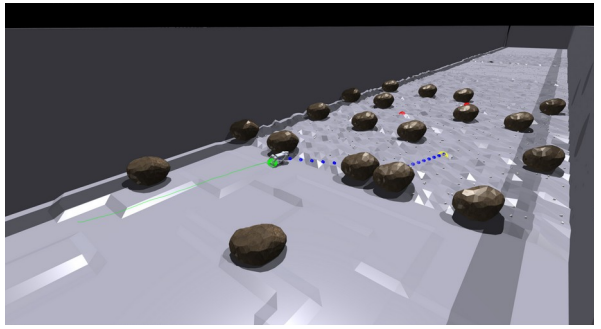
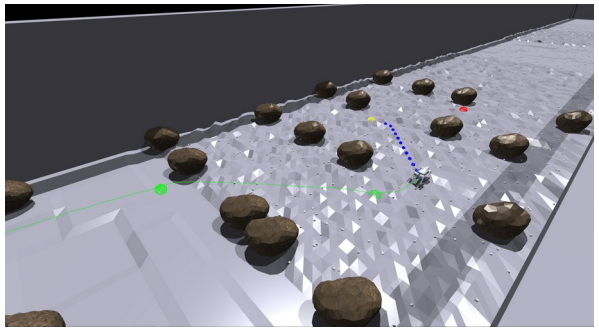
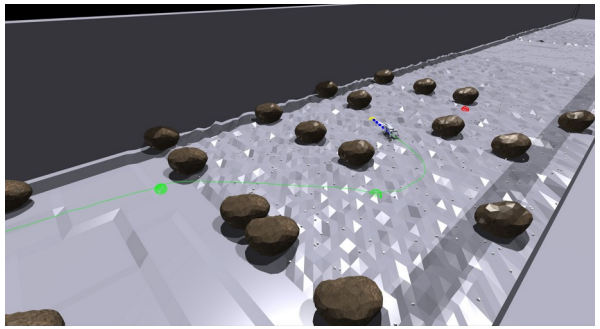
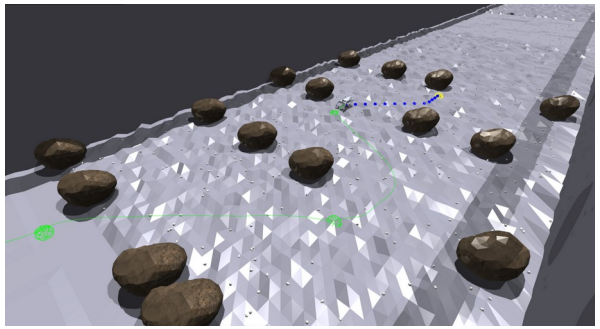
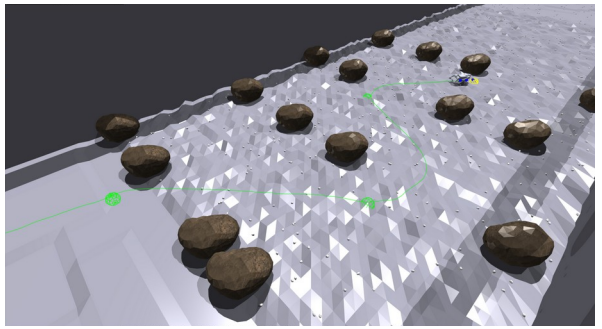
(b) Waypoint A — Waypoint B

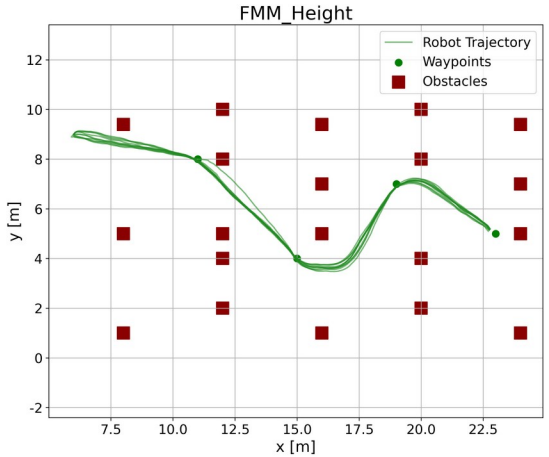


(c)Waypoint B — Waypoint C

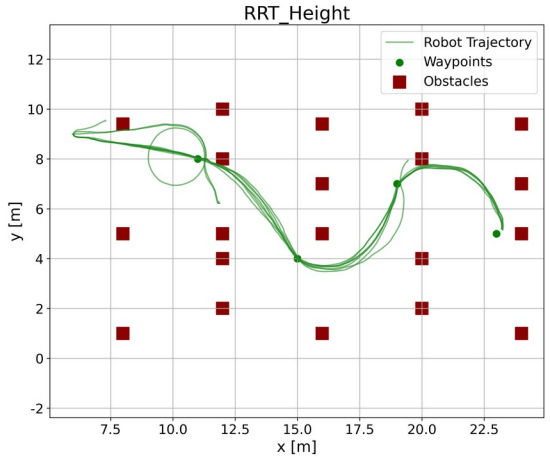


(d)Waypoint C — End Point

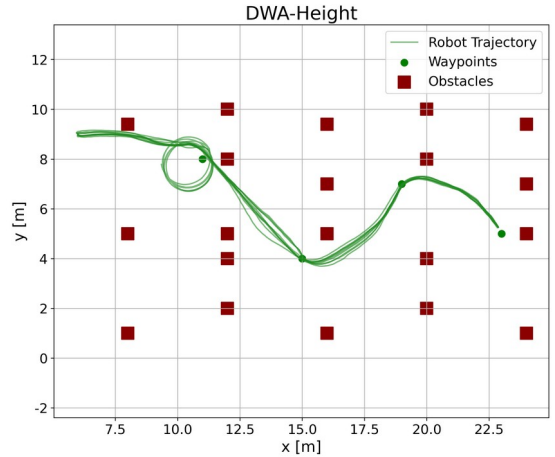










FMM_Height
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

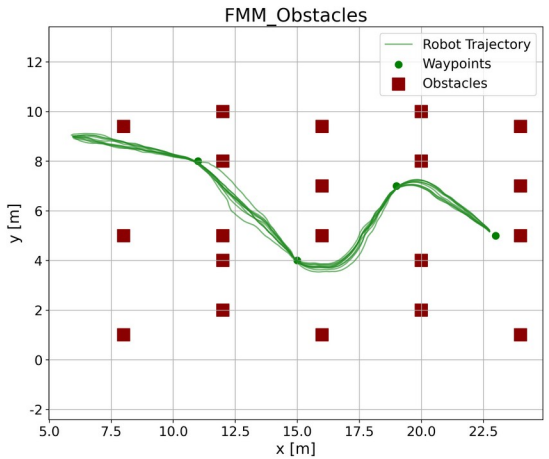
(a)


RRT_Height
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

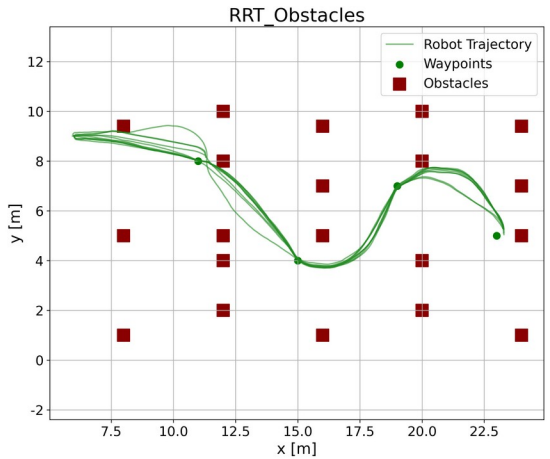
(b)


DWA-Height
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

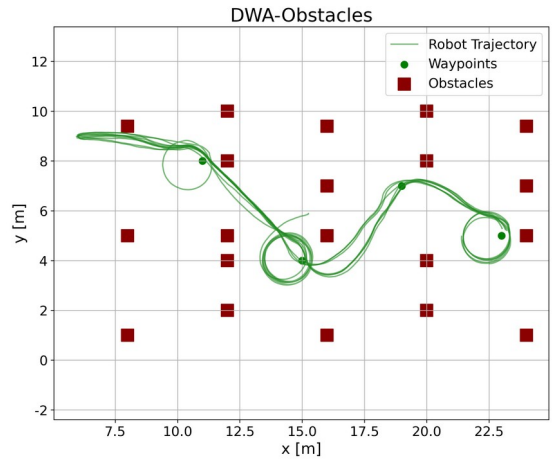
(c)


FMM_Obstacles
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

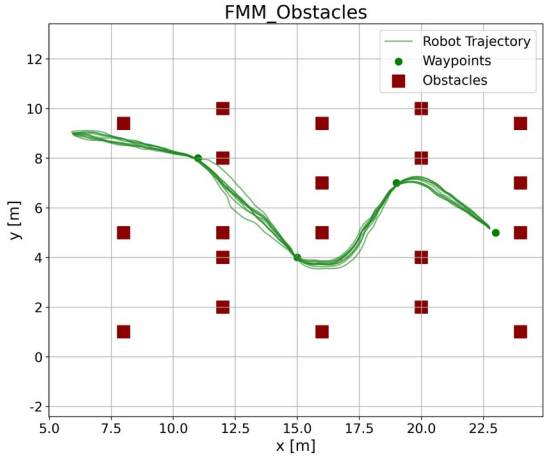
(a)


RRT_Obstacles
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

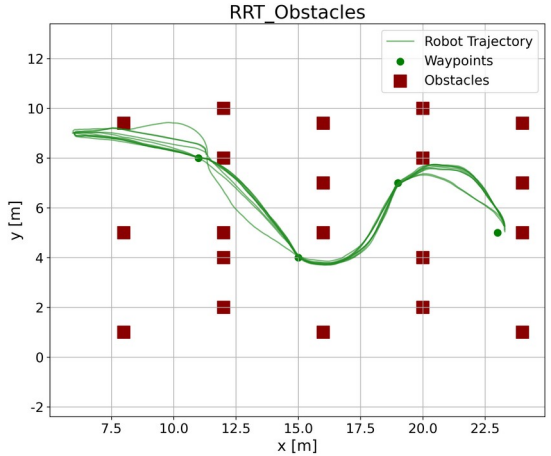
(b)


DWA-Obstacles
Robot Trajectory
Waypoints
Obstacles
y [m]
x [m]

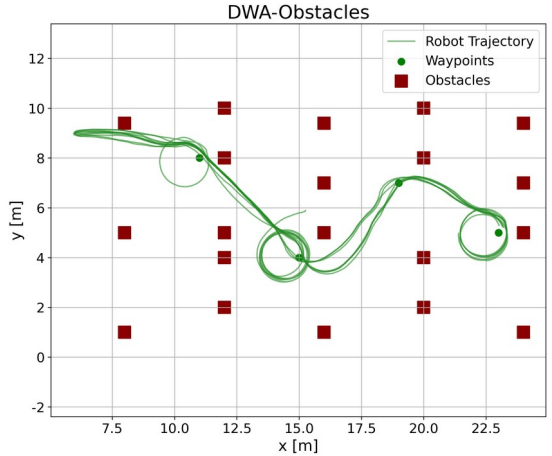
(c)


FMM_Obstacles
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

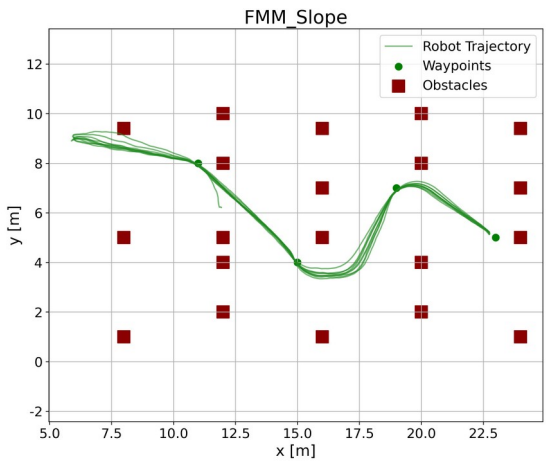
(a)


RRT_Obstacles
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

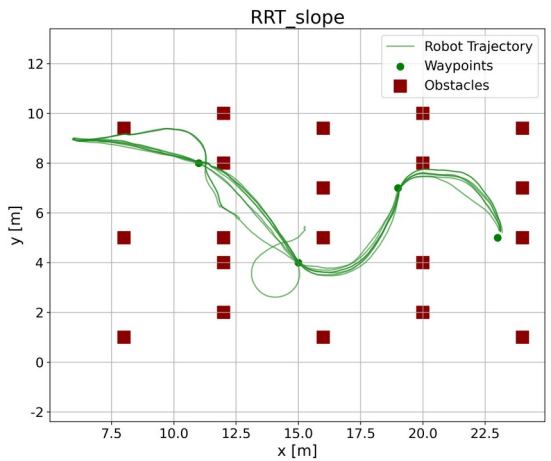
(b)


DWA-Obstacles
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

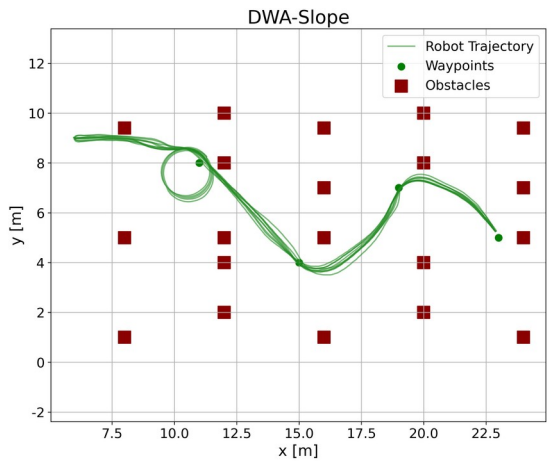
(c)


FMM_Slope
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

(a)


RRT_slope
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

(b)


DWA-Slope
Robot Trajectory
Waypoints
Obstacles
x [m]
y [m]

(c)